

RETHINKING THE EVALUATION AND OPTIMIZATION OF LLM-BASED SOCIAL SIMULATION*

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ABSTRACT

LLM-based social simulation is a promising complement to traditional social science methods such as surveys and behavioral experiments. A core question in this area is how to evaluate the fidelity of LLM-simulated human behavior and, in turn, how to optimize LLMs toward it. Prevailing practice evaluates by accuracy, checking whether the model selects the single response observed from a human, and accordingly trains the LLM to reproduce this one hard label. However, human behavior is inherently subjective: the same person in the same situation may reasonably act in different ways, so an observed response is only one draw from an underlying response distribution, which renders accuracy-based evaluation unreliable and hard-label training misleading. To address these problems, in this paper, we first introduce the *subjectivity coefficient*, an entropy-based quantity that distinguishes objective tasks such as coding from subjective tasks such as social simulation, and then use it to systematically analyze how accuracy-based evaluation and hard-label training fail as subjectivity grows. Based on the subjectivity coefficient, we further propose *Subjectivity-Adaptive soft-Label Training* (SALT): it pools observed outputs from semantically nearby inputs into soft distributional labels, with an aggregation radius adapted to the estimated subjectivity of each input; in the near-objective limit the neighborhood shrinks, so SALT naturally falls back to standard single-label training. Moreover, since existing datasets record only single observed responses and thus cannot support distributional evaluation, we construct SUBSIM, a benchmark of 19,300 contexts covering 193 annotators and 100 subjective questions. Since real-world data typically provide only a single observation per input, our experiments train models from single observed outputs while evaluating them against the full response distributions, thereby verifying the feasibility of our method in realistic settings. Extensive results on SUBSIM demonstrate the advantages of our method.

1 INTRODUCTION

LLM-based social simulation holds great potential for social science research, offering a low-cost, controllable, and scalable way to study problems such as opinion dynamics, political polarization, market behavior, and policy interventions (Lu et al., 2025; Wang et al., 2025b; Gao et al., 2024; Hua et al., 2023; Park et al., 2023; Sasahara et al., 2021; Bail et al., 2018). The core question in this area is how to evaluate the fidelity of LLM-simulated behavior (Dillion et al., 2023; Dominguez-Olmedo et al., 2024; Hu et al., 2026; Wang et al., 2025a; Samuel et al., 2025), and then, under the established criterion, how to train the model toward it. Most existing work approaches this question by directly following the practice of objective domains such as coding and math: collect one response for each person on each question, fine-tune the model to reproduce that response as the correct label, and check with accuracy-style metrics whether the model picks the same answer.

For objective tasks this recipe is perfectly sound: there is a well-defined correct answer, so a single label fully specifies the target, reproducing the label is exactly the desired behavior, and accuracy faithfully measures success. However, social simulation breaks this premise. Human behavior is stochastic rather than deterministic (Fleeson, 2001; Camerer, 1997; Slovic, 1995; Zaller, 1992; Ratcliff,

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1978; McFadden, 1974; Luce et al., 1959): the same person may act differently across comparable occasions, and people with similar profiles may choose different actions. An observed response is therefore only one draw from an underlying response distribution. A single label no longer specifies the target. During training, reproducing it collapses the model onto one action; at test time, whether the model hits that answer depends partly on chance rather than on how faithful the simulation is.

To study the above problems systematically, we first define the *subjectivity coefficient*, an entropy-based measure of the inherent randomness of human responses in a given decision context; under this definition, objective tasks (e.g., coding) have a near-zero subjectivity coefficient, while subjective tasks (e.g., social simulation) have a much higher one. Based on the subjectivity coefficient, we then formally show that the common recipe of accuracy evaluation plus single-label training breaks down as subjectivity grows. For evaluation, the recorded answer is itself only one random draw from the response distribution, so whether the model matches it reveals little about fidelity; for training, fitting the single label pulls the model away from the true response distribution. Finally, we propose *Subjectivity-Adaptive soft-Label Training* (SALT). For each context, SALT collects the answers observed at semantically similar contexts that share the same options and turns them into a soft label, and the neighborhood size automatically adapts to how subjective the context is estimated to be. When a context is nearly objective, the neighborhood shrinks and SALT falls back to standard single-label training, so hard-label supervision is recovered as the low-subjectivity special case.

Evaluating distributional fidelity requires reference human response distributions, but existing social-simulation datasets typically record only a single response for each context. We therefore construct SUBSIM, a benchmark in which 193 annotators answer 100 subjective survey questions, yielding 19,300 annotator-question pairs. Each pair carries an elicited response-propensity distribution obtained through probability-ball allocation. During training, models never see these distributions and receive only one derived hard action per pair; at test time, the full distributions are used to measure more accurately how faithfully the model reproduces human behavior. With Qwen3-8B as the backbone, SALT reduces aggregate KL divergence over standard supervised fine-tuning (SFT) by 77.6%, JSD by 45.9%, TVD by 31.2%, and MMD by 52.7%, with the largest gains in mid- and high-subjectivity settings, where one observed response reveals the least about the full distribution.

In summary, this paper makes contributions at three levels. (i) At the problem level, we identify why the prevailing accuracy-based evaluation and training practice is unreasonable for social simulation, and rigorously analyze this failure by introducing the subjectivity coefficient. (ii) At the method level, we propose SALT, which aggregates observations from semantically similar contexts into soft distributional labels with a subjectivity-adaptive radius. (iii) At the data level, we construct SUBSIM, a benchmark of 19,300 annotator-question pairs with human-annotated response distributions, and extensive experiments on it validate the effectiveness of SALT.

2 PRELIMINARIES

2.1 FORMULATION OF SOCIAL SIMULATION

We describe each individual in the target population by a persona vector u , which encodes demographic features, personality traits, prior beliefs, or other personal attributes. A decision context is the tuple $x := (u, s)$, where s is a situational description; we write $\mathcal{X}$ for the space of all decision contexts. At context x , the individual chooses from a candidate action set $\mathcal{A}(x) = \{a^{(1)}, \dots, a^{(K)}\}$, where each action is a natural-language description and $K = |\mathcal{A}(x)|$ may vary across contexts. For example, u may describe a 26-year-old graduate student living in a big city, s may ask which factor matters most when choosing a job, and the corresponding action set is $\mathcal{A}(x) = \{\text{“salary”}, \text{“stability”}, \text{“personal interest”}, \text{“work–life balance”}\}$. In another example, u may describe a retired teacher living in a small town, s may ask how the individual mainly gets news, and $\mathcal{A}(x) = \{\text{“social media”}, \text{“television”}, \text{“news apps”}\}$. Social simulation aims to use an LLM to mimic how these individuals respond and behave.

2.2 THE IDEAL OPTIMIZATION TARGET

Defining the ideal optimization target requires first understanding the nature of human behavior. Many social science theories have shown that human behavior is not deterministic but contains an inherent random component. Random utility theory (Luce et al., 1959; McFadden, 1974), stochastic

evidence-accumulation models (Ratcliff, 1978), and preference construction research (Slovic, 1995) all model a person’s choice as a draw from a probability distribution rather than as a fixed answer. This randomness is easy to see in daily life: the same person may pick a different dish from the same menu on different days, answer the same survey question differently when asked twice (Zaller, 1992), cooperate in one round of an economic game and defect in the next (Camerer, 1997), or be talkative at one gathering and quiet at another (Fleeson, 2001). This stochasticity is part of the target itself rather than label noise. Ideally, the model should therefore capture the uncertainty of human behavior: for each context, it should reproduce not only which action is most likely, but the full probability mass assigned to all plausible actions. Formally, the target is the latent response-propensity distribution $\phi^*(\cdot | x) \in \Delta(\mathcal{A}(x))$, namely the action distribution that would be observed if the same context could be measured repeatedly under comparable conditions, where $\Delta(\mathcal{A}(x)) = \{(q_1, \dots, q_K) : q_k \geq 0, \sum_k q_k = 1\}$ is the probability simplex over the available actions and we write $p_k := \phi^*(k | x)$ for brevity. On the model side, an LLM with parameters θ induces an action-level distribution $\phi_\theta(\cdot | x) \in \Delta(\mathcal{A}(x))$, obtained by normalizing the generation probabilities of the K candidate actions, with $\phi_\theta(k | x)$ written analogously. The ideal goal of social simulation is to learn a model whose induced action distribution is close to ϕ^* across contexts:

$$\hat{\theta} = \arg \min_{\theta} \mathbb{E}_x [D(\phi^*(\cdot | x), \phi_\theta(\cdot | x))], \quad (1)$$

where D measures mismatch on the action simplex and can be instantiated as various distance functions, e.g., total variation $D_{\text{TV}}(P, Q) := \frac{1}{2} \sum_{k=1}^K |P_k - Q_k|$, cross entropy $D_{\text{CE}}(P, Q) := -\sum_k P_k \log Q_k$, or KL divergence $D_{\text{KL}}(P \| Q)$.

2.3 THE CURRENT ACCURACY-BASED PRACTICE

In practice, ϕ^* is not observable: each context is recorded only once, yielding a dataset $\mathcal{D} = \{(x^i, a(x^i))\}$ in which $a(x) \in \mathcal{A}(x)$ is the single observed action, either one stochastic draw from $\phi^*(\cdot | x)$ or a single-action proxy derived from it. Let $j(x)$ denote the index of $a(x)$ and $\delta_{a(x)}$ the one-hot distribution that puts all mass on $a(x)$. Prevailing practice trains and evaluates against this single observation. On the training side, the model is trained to maximize the probability it assigns to the observed action:

$$\tilde{\theta} = \arg \max_{\theta} \mathbb{E}_x [\log \phi_\theta(j(x) | x)] = \arg \min_{\theta} \mathbb{E}_x [D_{\text{CE}}(\delta_{a(x)}, \phi_\theta(\cdot | x))], \quad (2)$$

which is exactly objective (1) with the unobservable target $\phi^*(\cdot | x)$ replaced by the one-hot vector $\delta_{a(x)}$ and D taken as the cross entropy. On the evaluation side, accuracy checks whether the model’s top-probability action coincides with the same single observation:

$$\text{Acc}(\theta) = \mathbb{E}_x [\mathbf{1}[\arg \max_k \phi_\theta(k | x) = j(x)]], \quad (3)$$

replacing the divergence in (1) by a 0–1 comparison between the model’s mode and the single draw. A natural question arises: how much do (2) and (3) actually capture of the ideal objective (1)? In the next section, we answer this question by rigorously analyzing the gap between them.

3 MISALIGNMENT ANALYSIS OF THE ACCURACY-BASED PRACTICE

3.1 EVALUATION METRIC ANALYSIS

Ideally, an evaluation metric should be consistent with true model quality: a model that scores better under the metric should also be closer to the true response distribution under the ideal objective (1). We now check whether accuracy (3) has this property.

Consider ranking two models θ_1 and θ_2 by accuracy at a context x with observed index $j = j(x)$, and let k_1 and k_2 denote the two models’ top actions. There are three cases.

(i) $k_1 = k_2$: the two models receive the same score. However, the true error $D_{\text{TV}}(\phi_{\theta_i}, \phi^*)$ depends on all K probability values rather than the top action alone, and the hidden gap can be nearly maximal. Suppose θ_1 matches the target exactly, $\phi_{\theta_1} = \phi^*$, while θ_2 puts all its mass on the shared top action k_1 , which is then the mode of ϕ^* with $p_{k_1} = p_{\max}$. Their true errors are

$$D_{\text{TV}}(\phi_{\theta_1}, \phi^*) = 0, \quad D_{\text{TV}}(\phi_{\theta_2}, \phi^*) = \frac{1}{2} \left[(1 - p_{\max}) + \sum_{k \neq k_1} p_k \right] = 1 - p_{\max}, \quad (4)$$

which differ by $1 - p_{\max}$ and reach $1 - 1/K$ under uniform behavior, yet accuracy scores the two models identically.

(ii) $k_1 \neq k_2$ and neither equals j : both models score zero, and accuracy again cannot tell them apart. As in case (i), consider a pair of models where one is perfect and the other is a point-mass model. The observed action j is one random draw from ϕ^* and may well miss the most probable action, so even the perfect model $\phi_{\theta_1} = \phi^*$, whose top action k_1 is the most probable one, can fall into this case, while θ_2 puts all its mass on another unobserved action, $\phi_{\theta_2} = \delta_{k_2}$. Their true errors are

$$D_{\text{TV}}(\phi_{\theta_1}, \phi^*) = 0, \quad D_{\text{TV}}(\phi_{\theta_2}, \phi^*) = \frac{1}{2} \left[(1 - p_{k_2}) + \sum_{k \neq k_2} p_k \right] = 1 - p_{k_2}, \quad (5)$$

so a perfect model and a model with error $1 - p_{k_2}$ receive the same zero score.

(iii) $k_1 \neq k_2$ and exactly one equals j , say $k_1 = j$: accuracy ranks θ_1 higher, and this is the only case where it expresses a preference. However, the preference can be exactly backward. Suppose θ_2 is the perfect model, $\phi_{\theta_2} = \phi^*$, with its top action k_2 the most probable one, while θ_1 puts all its mass on the draw $j \neq k_2$. Their true errors are

$$D_{\text{TV}}(\phi_{\theta_1}, \phi^*) = \frac{1}{2} \left[(1 - p_j) + \sum_{k \neq j} p_k \right] = 1 - p_j > 0 = D_{\text{TV}}(\phi_{\theta_2}, \phi^*), \quad (6)$$

so whenever the single draw misses the most probable action, accuracy prefers a strictly worse model over the perfect one, and the reversed gap $1 - p_j$ grows as behavior becomes more diffuse.

In summary, the above counterexamples show that accuracy is not a reliable measure of behavioral fidelity: models with the same score can differ substantially in true quality, and in the worst case accuracy even prefers a strictly worse model over the perfect one. A better accuracy score therefore does not imply a model closer to the true response distribution.

3.2 TRAINING OBJECTIVE ANALYSIS

We next examine the training objective (2). Its empirical signal at each training context is the one-hot target $\delta_{a(x)}$: the objective is monotonically increasing in the probability assigned to the observed action, so its optimum drives $\phi_{\theta}(j | x) \rightarrow 1$ on the training context, with no gradient signal that rewards distributing mass across multiple actions.

Proposition 1 (Single-observation training concentrates on one-hot labels). *Consider a finite training set in which each context x^i is observed once with hard label j^i . If the closure of the model-induced action distributions contains the probability simplex independently at each training context (an idealized full-capacity condition), then any empirical-risk minimizer $\hat{\theta}$ of objective (2) in this closure assigns $\phi_{\hat{\theta}}(k | x^i) = \mathbf{1}[k = j^i]$ for every training context x^i . In other words, the closer a model is trained toward the minimum loss, the closer its predicted distribution comes to putting probability one on the single observed answer at each training context.*

The proof is given in Section C.1. Under this point-mass solution, the total-variation error at a training context is:

$$D_{\text{TV}}(\phi_{\hat{\theta}}, \phi^*) = \frac{1}{2} \sum_k |\mathbf{1}[k = j] - p_k| = 1 - p_j, \quad (7)$$

which equals the total probability mass that ϕ^* assigns to actions other than the one observed. Note that the observed answer j is just one random draw: it tends to be a likely option but is not always the most likely one, so the error $1 - p_j$ is at least $1 - p_{\max}$. Even in the most favorable case where the draw hits the most likely option, diffuse behavior forces every probability, including the largest, to be small, so the error grows with subjectivity and reaches $(K - 1)/K$ at the uniform distribution.

The analysis above indicates, qualitatively, that the error of the single-label optimum is ultimately governed by how diffuse the response distribution is. To describe this relation quantitatively, we introduce a scalar that summarizes the dispersion of the entire distribution.

Definition 1 (Subjectivity coefficient). *The subjectivity coefficient of a decision context x is the negentropy of the target distribution:*

$$\kappa(x) := \sum_{k=1}^K p_k \log p_k \in [-\log K, 0], \quad (8)$$

where $p_k = \phi^*(k | x)$.

The coefficient equals zero when ϕ^* is a point mass (fully deterministic behavior) and $-\log K$ when it is uniform (maximum ambiguity). Throughout, we mainly work with the normalized subjectivity coefficient

$$s(x) := -\frac{\kappa(x)}{\log K} = \frac{H(\phi^*(\cdot | x))}{\log K} \in [0, 1], \quad (9)$$

where H is the Shannon entropy, so larger $s(x)$ means more subjective behavior. Crucially, s is a property of the decision context x , not of any model: it characterizes how inherently subjective a given scenario is. Under this view, objective tasks such as coding and math occupy the near-zero- s regime, subjective simulation tasks span the full range, and different task types differ only in their degree of subjectivity along this common axis. Entropy is the standard measure of how uncertain a distribution is, and it takes all K probabilities into account rather than a single one. We now show that the training error grows with s through an explicit lower bound.

Proposition 2 (Subjectivity lower-bounds the error of single-label training). *Let $p_{\max} = \max_k p_k$, $s = s(x)$, and $K \geq 3$. Any point-mass model incurs training error*

$$D_{\text{TV}}(\phi_\theta, \phi^*) \geq 1 - p_{\max} \geq \frac{s \log K - \log 2}{\log(K-1)}, \quad (10)$$

where the first inequality follows from Equation (7) since $p_j \leq p_{\max}$, and the second follows from Fano's inequality (Appendix C.2). Hence the more subjective the context, i.e., the larger s , the larger the unavoidable error of any point-mass model; the lower bound is approximately s for large K , so the unavoidable error is at least roughly the normalized subjectivity of the context. At the uniform distribution, where $s = 1$, the exact error $1 - p_{\max}$ equals $1 - 1/K$.

The essential cause behind the failures on both the evaluation and the training side is the same: one observed answer per context carries too little information about a diffuse response distribution. Ideally, if each context were annotated many times, the empirical answer frequencies would recover the true distribution and both problems would disappear. This is unrealistic, however: reliable model ranking alone would require $\Omega(K^{2s})$ repeated observations per context (Appendix C.3), while real data provide exactly one. A classical alternative is to directly merge the observed answers of similar contexts, as in local smoothing (Nadaraya, 1964; Watson, 1964). Merging, however, involves a trade-off: pooling more neighbors supplies more distributional information, but neighbors are only similar rather than identical, so pooling also mixes in answers from different distributions and biases the target. In the next section, we analyze this trade-off theoretically and derive an algorithm with a context-adaptive merging radius.

4 SUBJECTIVITY-ADAPTIVE SOFT-LABEL TRAINING (SALT)

SALT replaces each one-hot target with a soft label aggregated from semantically similar contexts, and chooses the neighborhood size to balance the trade-off above. Concretely, given $\mathcal{D} = \{(x^i, a(x^i))\}_{i=1}^n$, SALT outputs a soft label $\hat{\phi}(\cdot | x)$ for each context. Because actions are comparable only within a shared candidate set, we first partition contexts into action-space groups:

$$\mathcal{G}_g = \{x \in \mathcal{X} : \mathcal{A}(x) = \mathcal{A}_g\}, \quad g = 1, \dots, G, \quad (11)$$

where $\mathcal{A}(x)$ is the candidate action set of context x defined in Section 2.1 and $\mathcal{A}_g$ is the shared candidate set of group g ; in survey data, for instance, all contexts answering the same question with the same options form one group. Within each group, contexts are embedded with a pretrained encoder and compared by ℓ_2 distance d . The neighborhood of $x \in \mathcal{G}_g$ is

$$\mathcal{N}(x) = \{x' \in \mathcal{G}_g : d(x', x) \leq r(x)\}, \quad (12)$$

where the radius $r(x)$ is adaptive; how to choose it is the key design question, addressed below. Since all neighbors share $\mathcal{A}_g$, their hard actions define a local empirical distribution:

$$\hat{\phi}(k | x) := \frac{1}{|\mathcal{N}(x)|} \sum_{x' \in \mathcal{N}(x)} \mathbf{1}[a(x') = a^{(k)}], \quad k = 1, \dots, K, \quad (13)$$

and the model is trained to match this distributional target:

$$\mathcal{L}_\theta^{\text{agg}} = \sum_{x \in \mathcal{D}} D_{\text{KL}}(\hat{\phi}(\cdot | x) \parallel \phi_\theta(\cdot | x)), \quad (14)$$

where $\phi_\theta(\cdot | x) \in \Delta(\mathcal{A}(x))$ is obtained by normalizing generation probabilities across candidates. Note that this differs from prior grouping-based methods (Huang et al., 2026; Cao et al., 2025), which partition samples into disjoint groups (e.g., by demographic attributes) and let all samples in a group share one target distribution, whereas SALT centers a neighborhood at each context, so every context receives its own soft label.

The key remaining question is how to determine the radius $r(x)$. To answer it, we bound the error of aggregation within one action-space group $\mathcal{G}_g$, writing $d_\mathcal{X}$ for the intrinsic dimension of its context space, $n_g = |\mathcal{G}_g|$ for the number of contexts in the group, and K for the number of actions. The dimension enters through the volume of a neighborhood: a ball of radius r in a $d_\mathcal{X}$ -dimensional space holds a fraction $\asymp r^{d_\mathcal{X}}$ of the contexts, so $|\mathcal{N}(x)| \asymp n_g \cdot r^{d_\mathcal{X}}$. The bound rests on a smoothness assumption that formalizes the intuition behind SALT, namely that semantically nearby contexts induce similar response distributions; it is stated as an L -Lipschitz condition in the embedding distance d (Assumption 1 in Appendix C.4). The statistical part of the bound depends on how many actions carry substantial probability, captured by the effective action count.

Definition 2 (Effective number of actions). *For a context x , the effective number of actions is*

$$K_{\text{eff}}^*(x) := \left(\sum_{k=1}^K \sqrt{\phi^*(k | x)} \right)^2 \in [1, K], \quad (15)$$

which attains its lower endpoint when $\phi^(\cdot | x)$ is a point mass and its upper endpoint when $\phi^*(\cdot | x)$ is uniform. It further satisfies $K_{\text{eff}}^*(x) \geq K^{s(x)}$, so it increases with the subjectivity of the context; this bound and the remaining properties used below are established in Remark 1.*

Combining the decomposition with statistical and bias bounds yields the main tradeoff.

Theorem 1 (Aggregation–Estimation Tradeoff). *Suppose ϕ^* is L -Lipschitz in the embedding distance d (Appendix C.4). With $|\mathcal{N}(x)| \asymp n_g \cdot r^{d_\mathcal{X}}$, where $\asymp (\lesssim)$ denotes equality (inequality) up to constant factors, let $\varepsilon_{\text{opt}} := \mathbb{E}[D_{\text{TV}}(\phi_\theta(\cdot | x), \hat{\phi}(\cdot | x))]$. The expected error satisfies*

$$\mathbb{E}[D_{\text{TV}}(\phi_\theta(\cdot | x), \phi^*(\cdot | x))] \lesssim \underbrace{\varepsilon_{\text{opt}}}_{\text{opt. error}} + \underbrace{L \cdot r}_{\text{bias}} + \underbrace{\sqrt{\frac{K_{\text{eff}}^*(x)}{n_g \cdot r^{d_\mathcal{X}}}}}_{\text{stat. error}}. \quad (16)$$

The bias increases in r while the statistical error decreases in r (since $|\mathcal{N}(x)|$ grows with $r^{d_\mathcal{X}}$). When the aggregation loss is optimized so that ε_{opt} is negligible, balancing the remaining two terms gives the bias–variance optimized radius and the corresponding optimized upper bound:

$$r^* \asymp \left(\frac{K_{\text{eff}}^*(x)}{n_g L^2} \right)^{1/(d_\mathcal{X}+2)}, \quad \mathbb{E}[D_{\text{TV}}] \lesssim \varepsilon_{\text{opt}} + \left(\frac{L^{d_\mathcal{X}} \cdot K_{\text{eff}}^*(x)}{n_g} \right)^{1/(d_\mathcal{X}+2)}. \quad (17)$$

In the worst case ($K_{\text{eff}}^(x) = K$), the bias–statistical term becomes $(L^{d_\mathcal{X}} K / n_g)^{1/(d_\mathcal{X}+2)}$.*

The proof is given in Section C.6. The result has two implications. First, with negligible optimization error, oracle aggregation improves as n_g grows, whereas a local point-mass fit still incurs error at least $1 - p_{\text{max}}$ on the context (Equation (10)). Second, r^* increases with $K_{\text{eff}}^*(x)$: the more subjective a context, the larger its neighborhood should be.

Practical implementation of the oracle radius. Theorem 1 gives an oracle radius $r^* \asymp (K_{\text{eff}}^*(x)/(n_g L^2))^{1/(d_\mathcal{X}+2)}$. Since $K_{\text{eff}}^*(x)$ is defined through $\phi^*(\cdot | x)$, it is unobserved in training. In practice we evaluate the *same* expression at the model’s current distribution, which gives the model-based estimate $\hat{K}_{\text{eff}}(x) := (\sum_k \sqrt{\phi_\theta(k | x)})^2$. This is reasonable for two reasons. The aggregation loss (Equation (14)) is itself a distribution-matching objective, so it drives $\phi_\theta(\cdot | x)$ toward $\phi^*(\cdot | x)$, and the expression is continuous in the distribution, so the estimate returns to the true count as the optimization error vanishes. Exact recovery is moreover not required, because $r(x)$ depends on the count only through the power $1/(d_\mathcal{X} + 2)$, so a multiplicative error in the count is damped into a much smaller relative change in the radius. Remark 2 makes both precise. The Lipschitz constant L and other constants are absorbed into a tunable hyperparameter C , giving

$$r(x) = C \cdot \left(\frac{\hat{K}_{\text{eff}}(x)}{n_g} \right)^{1/(d_\mathcal{X}+2)}. \quad (18)$$

The estimate is refreshed periodically during training. This substitution preserves the oracle bound’s monotone dependence on action dispersion, and we evaluate it through fixed- K_{eff} , fixed-neighborhood, and radius-scale ablations. The theory thus does not merely justify aggregation; it specifies when to aggregate more: contexts with diffuse predicted behavior require larger neighborhoods to reduce statistical error, while concentrated contexts should remain close to their observed hard label to avoid unnecessary smoothing. In the zero-subjectivity limit, $\hat{K}_{\text{eff}}(x) \rightarrow 1$ and the radius contracts toward its minimum, so the soft label concentrates on the observed action and SALT approaches standard hard-label training. Hard-label supervision is therefore recovered as the low-subjectivity special case of SALT. More details and the complete algorithm can be found in Appendix D.

5 EXPERIMENTS

5.1 SUBJSIM BENCHMARK

Evaluating distributional fidelity requires the true response distributions as ground truth, which existing single-response datasets cannot provide. We therefore construct SUBJSIM, where 193 annotators first answer 30 demographic questions that define their persona vectors and then annotate 100 subjective survey questions, yielding 19,300 persona-question contexts. For each context, the annotator allocates probability balls across the candidate options (Van Lenthe, 1993; Delavande et al., 2010), producing a response-propensity distribution that is used only for evaluation. All training methods, including SALT’s aggregation, see just one answer per context: the option receiving the largest share of that annotator’s balls, with ties resolved in favor of the lowest-indexed option. The questions are organized into eight topic domains: economy, politics, technology, social issues, culture, health, environment, and education. Their subjectivity coefficients cover the full range from near-deterministic to near-uniform, enabling evaluation across low, mid, and high subjectivity regimes (see Figure 12; construction details in Appendix G).

5.2 EXPERIMENTAL SETUP

We train on the samples of about 85% of the respondents and test on those of the remaining ones, which gives 16,500 training pairs and 2,800 test pairs. All experiments use Qwen3-8B (Yang et al., 2025) as the backbone. We compare SALT with the untrained zero-shot backbone, SFT, DPO, PPO, and DSA (Huang et al., 2026). DPO constructs preference pairs and PPO derives reward signals from the hard labels described in Section 5.1, so that all methods observe identical training information. DSA is the most recent distribution-level baseline, which fine-tunes the LLM to match the response distributions of demographic groups and to align distribution shifts across groups. At test time, models output distributions over candidate options via generation-probability normalization. We report KL, JSD, TVD, and linear-kernel MMD computed on the full test set; see Appendix D for further experimental settings.

5.3 MAIN RESULTS

The main results are shown in Table 1. The pretrained backbone performs poorly, and neither DPO nor PPO improves over it; DPO in fact degrades KL substantially. We attribute this to the construction of the preference signal from a single hard action: the observed action serves as the positive and the remaining options as negatives, yet under subjective behavior these options may themselves carry substantial probability, so responses the respondent might well choose are suppressed as negatives, making the resulting signal even noisier than fitting the label directly. SFT is the strongest baseline, suggesting that directly fitting the observed labels already captures a substantial part of the underlying behavioral structure. DSA ranks between the preference-based methods and SFT, clearly outperforming the former, which confirms the benefit of distribution-alignment training. However, DSA groups respondents by only a few discrete background attributes, and all individuals within a group share a single target distribution; the supervision is therefore noisy at the individual level, leaving DSA behind SFT. SALT controls this noise with its adaptive radius, pooling only sufficiently similar contexts while still collecting enough answers for a reliable distribution estimate. It performs best on all four metrics in every domain, and over the full test set it reduces KL over SFT by 77.6%, JSD by 45.9%, TVD by 31.2%, and MMD by 52.7%. A paired bootstrap over test contexts confirms that all of these improvements are statistically significant (Appendix F).

Table 1: Main results on SubjSim. All metrics are divergences (lower is better). Best values are in **bold** on a blue background; second-best values are underlined on a sand background.

Method	Economy				Political				Technology			
	KL↓	JSD↓	TVD↓	MMD↓	KL↓	JSD↓	TVD↓	MMD↓	KL↓	JSD↓	TVD↓	MMD↓
Pretrained	4.6285	0.2851	0.5735	0.6346	4.1453	0.3044	0.6206	0.6586	4.2015	0.2547	0.5335	0.5502
SFT	1.3543	0.1542	0.3951	0.3283	1.2865	0.1568	0.4145	0.3167	1.3409	0.1537	0.3956	0.3215
DPO	6.9962	0.2752	0.5573	0.6049	7.2926	0.2834	0.5816	0.5861	6.9174	0.2580	0.5305	0.5450
PPO	4.0161	0.2672	0.5495	0.5863	4.4698	0.2726	0.5724	0.5684	4.2959	0.2502	0.5251	0.5310
DSA	3.2032	0.2332	0.4920	0.4337	3.4679	0.2442	0.5161	0.4436	3.4744	0.2337	0.4888	0.4275
SALT (Ours)	0.3129	0.0892	0.2839	0.1640	0.2535	0.0724	0.2473	0.1247	0.2817	0.0815	0.2736	0.1466
Method	Social				Culture				Health			
	KL↓	JSD↓	TVD↓	MMD↓	KL↓	JSD↓	TVD↓	MMD↓	KL↓	JSD↓	TVD↓	MMD↓
Pretrained	4.2286	0.2720	0.5562	0.6079	3.0456	0.2445	0.5175	0.5646	4.7412	0.2822	0.5737	0.6467
SFT	1.3259	0.1588	0.3977	0.3354	0.8578	0.1209	0.3459	0.2739	1.5169	0.1566	0.3886	0.3286
DPO	6.8092	0.2608	0.5289	0.5629	5.6298	0.2385	0.5050	0.5398	6.5058	0.2501	0.5152	0.5366
PPO	4.2388	0.2524	0.5210	0.5471	3.5294	0.2278	0.4909	0.5180	4.4975	0.2409	0.5059	0.5215
DSA	3.2049	0.2334	0.4963	0.4537	4.1157	0.2623	0.5259	0.5762	3.8155	0.2433	0.4889	0.4561
SALT (Ours)	0.3168	0.0905	0.2863	0.1656	0.2593	0.0746	0.2652	0.1548	0.2995	0.0859	0.2793	0.1601
Method	Environment				Education				ALL			
	KL↓	JSD↓	TVD↓	MMD↓	KL↓	JSD↓	TVD↓	MMD↓	KL↓	JSD↓	TVD↓	MMD↓
Pretrained	5.3499	0.3091	0.6199	0.6506	6.1097	0.3662	0.6842	0.8295	4.2801	0.2851	0.5745	0.6350
SFT	0.8800	0.1373	0.3837	0.2685	2.0897	0.2324	0.5246	0.4818	1.2871	0.1524	0.3953	0.3194
DPO	7.2403	0.3099	0.6163	0.6573	8.4950	0.3276	0.6291	0.7074	6.8819	0.2690	0.5497	0.5774
PPO	3.5411	0.2914	0.6035	0.6206	5.1310	0.3337	0.6447	0.7245	4.1393	0.2588	0.5403	0.5578
DSA	3.0895	0.2116	0.4613	0.3496	2.7515	0.2260	0.4870	0.4152	3.3904	0.2343	0.4937	0.4418
SALT (Ours)	0.2983	0.0870	0.2705	0.1407	0.3866	0.1095	0.3229	0.1947	0.2880	0.0825	0.2720	0.1510

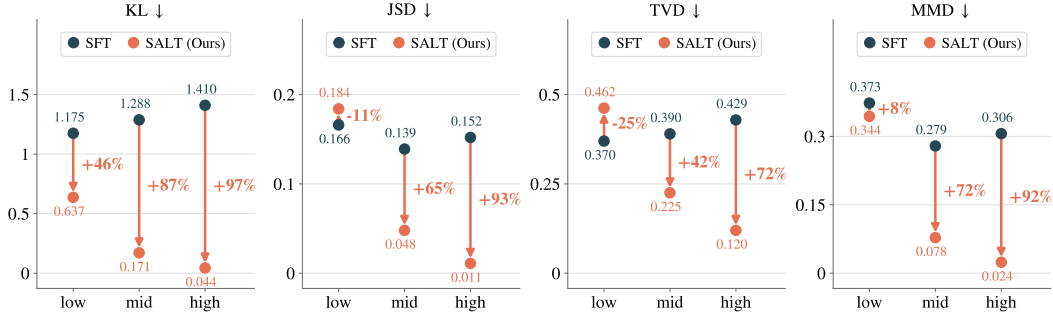


Figure 1: Subjectivity-stratified comparison between SFT and SALT on SUBJSIM, over low-, mid-, and high-subjectivity questions. All metrics are divergences (lower is better), and percentages give SALT’s relative change over SFT.

5.4 PERFORMANCE COMPARISON ACROSS SUBJECTIVITY LEVELS

A core design of SALT is that the aggregation is calibrated by the estimated subjectivity of each context, and we now examine experimentally whether this design indeed leads to better performance. We compute the normalized subjectivity coefficient $s(x)$ in (9) directly from the annotated response-propensity distributions. We then partition the questions into low, mid, and high strata at the 1/3 and 2/3 quantiles of $s(x)$ and compare SFT and SALT within each stratum (Figure 1). The results match the analysis in Section 3. SFT is competitive in near-deterministic contexts but degrades steadily as targets become diffuse, whereas SALT’s advantage grows with subjectivity; in the mid and high strata SALT outperforms SFT on all metrics, and in the high stratum it reduces KL by 96.9%. In the lowest-subjectivity regime the comparison is mixed, with SALT improving KL and MMD but falling behind SFT on JSD and TVD. This is what we would expect, because such questions have a clear majority answer, so there is little for aggregation to add and pooling neighbors can only blur a target that is already sharp.



Figure 2: Ablation over neighbor selection. Top- N varies N (teal, bottom axis); fixed K_{eff} substitutes a constant for $\hat{K}_{\text{eff}}(x)$ (navy, top axis). Both share the y axis and reference lines; SFT lies far above the plotted range and is reported as text.

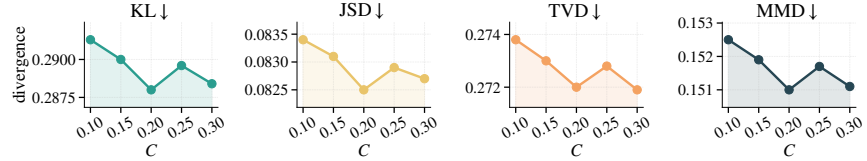


Figure 3: Sensitivity of SALT to the radius scale C (divergences, lower is better).

5.5 ABLATION STUDIES

The core design of SALT is the context-adaptive merging radius, so we ask whether such adaptivity is really necessary. We build three baselines that differ from SALT only in how neighbors are selected: Top- N gives every context the same number of nearest neighbors; fixed K_{eff} replaces $\hat{K}_{\text{eff}}(x)$ in the radius formula by one shared constant, so every context gets the same radius; Global-Freq drops similarity altogether and treats everyone who answered the same question as a neighbor.

From the results shown in Figure 2, we draw three conclusions. First, every variant improves substantially over SFT, which shows that aggregation is already useful on its own because it replaces a single observed answer with a distributional target. Second, how neighbors are selected matters: Top- N degrades as N grows and dissimilar contexts enter the neighborhood, fixed K_{eff} stops improving and stays worse than SALT on every metric, and the adaptive $\hat{K}_{\text{eff}}(x)$ is best on all four. Third, Global-Freq is clearly worse than SALT, so similarity is necessary: pooling everyone who answered the same question mixes dissimilar respondents and biases the soft label.

5.6 PARAMETER ANALYSIS

The main hyperparameter of SALT is the radius scale C in (18). The theory in Section 4 fixes only how the radius should scale with $\hat{K}_{\text{eff}}(x)$ and n_g ; the remaining constants, including the Lipschitz constant L , are absorbed into C , which is therefore chosen empirically. Figure 3 reports performance for $C \in \{0.10, 0.15, 0.20, 0.25, 0.30\}$. All metrics improve monotonically as C increases from 0.10 to 0.20: a larger radius admits more genuinely similar neighbors, so the aggregated soft labels become more reliable. Beyond 0.20 the improvement stops and fluctuates slightly, as a wider radius includes less similar contexts and mild oversmoothing offsets the gains; the trade-off is best around $C = 0.20$, which we use as the default. Importantly, the variation across the whole range is small—about 0.003 in KL and at most 0.002 in the other metrics—so SALT does not rely on a finely tuned radius constant.

6 CONCLUSION

In this paper, we studied how to evaluate and train LLMs for social simulation when human behavior is subjective. We introduced the subjectivity coefficient, which places objective and subjective tasks on a common axis and explains why accuracy-based evaluation and hard-label training become unreliable as subjectivity grows. We then proposed SALT, which turns each observed response into a soft distributional label aggregated over a subjectivity-adaptive neighborhood, and built SUBSIM, whose annotators provide full response distributions for 19,300 persona-question contexts, making distributional evaluation possible. SALT improves distributional alignment over all baselines, with the largest gains in high-subjectivity regimes. Future work includes extending this framework to sequential, multi-agent, and open-ended settings.

ETHICS STATEMENT

SUBJ_{SIM} involves human annotation. All 193 annotators participated voluntarily; before starting the annotation task, they were informed of the purpose of the study, the type of data collected, and their right to withdraw at any time. Annotation compensation was calculated so that the resulting hourly rate is guaranteed to exceed the highest applicable local hourly wage standard. To protect privacy, no directly identifying information (such as names or contact details) was collected, all responses were recorded under anonymized identifiers, and the released data contain only demographic attribute values and annotated response distributions with no link back to individuals. The survey questions themselves were screened for cultural suitability for the annotator population during dataset construction, and questions flagged as unsuitable were removed (Appendix G). The representativeness limits of our annotator pool are discussed in Appendix G.3.

AI USE STATEMENT

We used large language models only as writing and figure assistants in the preparation of this paper: ChatGPT was used to check grammar and correct typos in the manuscript, and Codex was used to assist in producing the figures. No LLM was used to generate research ideas, analyses, results, or claims. Separately, and as part of the research methodology itself rather than paper preparation, DeepSeek-chat was used to translate and pre-screen survey questions during the construction of SUBJ_{SIM}, as documented in Appendix G.

REPRODUCIBILITY STATEMENT

We have taken several measures to make our results reproducible. All theoretical results are stated with their assumptions in Sections 3 and 4, and complete proofs are given in Sections C.1 to C.6. The full SALT procedure is specified in Algorithm 1, with implementation details, the theory-to-implementation mapping (Table 2), and all training hyperparameters (Table 3) in Section D. The construction of SUBJ_{SIM}, including source surveys, screening and translation pipeline, the exact screening prompt, annotation protocol, and annotator demographics, is documented in Section G. Evaluation is deterministic, and the backbone model (Qwen3-8B), context encoder (Qwen3-embedding-8b), and training stack (LLaMA-Factory, DeepSpeed ZeRO-2 on 8×H20) are publicly available. We will release the SUBJ_{SIM} dataset, the annotation platform specification, and the training and evaluation code upon publication.

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A LIMITATIONS

SUBJ_{SIM} uses elicited response propensities rather than direct repeated-choice frequencies. The annotator-level split tests unseen personas within known questions rather than transfer to entirely new decision contexts. SALT also depends on behavioral smoothness in the representation space, so poor embeddings or discontinuous response patterns can make aggregation harmful.

B RELATED WORK AND POSITIONING

LLM agents for social simulation. Recent work uses LLM agents to simulate individuals, groups, and social systems, ranging from interactive generative agents to opinion dynamics, polarization, market behavior, and historical or policy simulations (Park et al., 2023; Gao et al., 2024; Hua et al., 2023; Sasahara et al., 2021; Bail et al., 2018). This line of work establishes that LLMs can produce plausible social behavior and can be embedded in multi-agent environments. Our focus is different: we ask what behavioral target such agents should be trained and evaluated against. Rather than treating simulation quality as matching a single observed response, we formulate the target as recovering the full distribution over plausible actions for a persona-context pair.

Single-action imitation and preference optimization. The dominant way to adapt LLM agents is supervised imitation of observed actions, often evaluated by point-prediction accuracy. Preference optimization methods such as DPO and PPO provide a stronger alignment toolkit by learning from chosen/rejected comparisons (Rafailov et al., 2023; Schulman et al., 2017). However, when the data provide only one realized action per context, both supervised and preference objectives still receive hard, mode-like supervision. They therefore learn which action was observed, or which action should be preferred to alternatives, rather than the entire behavioral frequency structure. This is the same hard-supervision gap that motivates our misalignment analysis, though our formal result focuses on the single-observation SFT objective.

Distributional evaluation and distribution alignment. A separate line of work changes the evaluation target: instead of reporting only accuracy, it compares model outputs to response distributions using KL, JSD, TVD, Wasserstein distance, or related survey-response divergences. DSA is closest to our work in target space: it also aims to match survey response distributions, but it does so through distribution-shift alignment rather than context-level hard-label aggregation (Huang et al., 2026). Our setting differs in the supervision assumption. DSA and related distributional alignment methods rely on distributional supervision that is available or can be constructed at the population/question level. SALT targets the stricter single-observation regime: it does not train on the persona-specific probability-ball distribution, but constructs distributional supervision by pooling hard observations from semantically similar contexts, following the repeated-measurement intuition in the main analysis.

Human label variation. NLP research on annotation has similarly argued that disagreement among annotators carries signal rather than noise, advocating soft labels, annotator-aware models, and evaluation against label distributions instead of a single gold label (Aroyo & Welty, 2015; Plank, 2022; Uma et al., 2021). These works concern classification-style NLP annotations and model variation at the label or annotator level. Our setting differs in both object and ground truth: the estimation target is a persona-conditional behavioral distribution for social simulation, and SUBJ_{SIM} measures it directly through probability-ball allocation rather than approximating it from a handful of discrete annotations.

Context aggregation and nonparametric estimation. SALT is related to nonparametric smoothing and kernel regression, where local neighborhoods are used to estimate a conditional response function (Nadaraya, 1964; Watson, 1964; Silverman, 2018). The key difference is the object being smoothed and the constraint imposed by social-simulation data. We aggregate over contexts that share an action space and are close in persona-context embedding space, producing a soft label over discrete actions from otherwise hard observations. The adaptive radius is not a generic hyperparameter search: it is derived from the subjectivity-controlled bias–variance tradeoff, so more diffuse behavioral distributions receive broader aggregation while nearly deterministic contexts remain local.

C THEORETICAL PROOFS

C.1 PROOF OF PROPOSITION 1 (SINGLE-OBSERVATION SFT CONCENTRATES ON EMPIRICAL ONE-HOT LABELS)

Proof. For each training context x^i , the empirical SFT contribution is $-\log \phi_\theta(j^i | x^i)$ at the action level. Over the action-distribution simplex, this term is minimized by maximizing $\phi_\theta(j^i | x^i)$, whose largest possible value is 1. Because the statement is made in the closure of the model-induced action distributions, this boundary point is included and can be selected independently for every training context, yielding $\phi_\theta(k | x^i) = \mathbf{1}[k = j^i]$ for all i . Since $-\log p$ is uniquely minimized at $p = 1$, every empirical-risk minimizer must saturate the per-context term at every training context, so any minimizer $\tilde{\theta}$ in this closure satisfies $\phi_{\tilde{\theta}}(k | x^i) = \mathbf{1}[k = j^i]$ for all i . For finite softmax parameterizations, the same boundary point may be approached only in the limit: any sequence with empirical risk approaching the infimum must have $\phi_\theta(j^i | x^i) \rightarrow 1$ and hence $\phi_\theta(k | x^i) \rightarrow 0$ for every $k \neq j^i$. This is an empirical-risk statement for finite single-observation data; with repeated observations from the same context, the population cross-entropy optimum would instead match the conditional label distribution. $\square$

C.2 PROOF OF THE FANO STEP IN PROPOSITION 2

Proof. Write $e := 1 - p_{\max}$ and $H := H(\phi^*(\cdot | x)) = -\kappa$. Fano's inequality for the error of the optimal single guess (the mode) states $H \leq h(e) + e \log(K - 1)$, where $h(\cdot)$ is the binary entropy function. Since $h(e) \leq \log 2$, we obtain $H \leq \log 2 + e \log(K - 1)$, and rearranging gives

$$1 - p_{\max} = e \geq \frac{H - \log 2}{\log(K - 1)} = \frac{-\kappa - \log 2}{\log(K - 1)} = \frac{s \log K - \log 2}{\log(K - 1)} \quad (19)$$

for $K \geq 3$, where the last step uses $s = -\kappa / \log K$ from Equation (9). Combining with $D_{\text{TV}}(\phi_\theta, \phi^*) \geq 1 - p_{\max}$ from Equation (7) yields Equation (10). The bound is vacuous when $s \log K < \log 2$, i.e., for weakly subjective contexts, which is consistent with the analysis: the failures of the accuracy-based pipeline concentrate in the high-subjectivity regime. For $K = 2$, Fano's inequality reduces to $H \leq h(e)$, giving $e \geq h^{-1}(H)$ instead; the conclusion that the error grows with subjectivity is unchanged. $\square$

C.3 PROOF OF PROPOSITION 3 (LOCAL LOWER BOUND UNDER NEAR-UNIFORM BEHAVIOR)

Proposition 3 (Local lower bound under near-uniform behavior). *Fix a K -action context and consider true distributions in a near-uniform neighborhood of radius $O(1/K)$, so that $e^{-2\kappa} = \Theta(K^2)$. There exist two alternatives P, Q in this neighborhood and two fixed point-mass models whose D_{TV} -error ranking is reversed under P and Q , while $D_{\text{KL}}(P \| Q) = O(1/K^2)$. Consequently, for any fixed $\eta > 0$, any procedure that identifies the correct ranking from n i.i.d. observations with error probability at most $\frac{1}{2} - \eta$ in this local problem requires $n = \Omega(K^2) = \Omega(e^{-2\kappa})$ samples.*

Proof. We give an explicit two-point construction. Let u be the uniform distribution on K actions and choose $\varepsilon = cK^{-3/2}$ for a sufficiently small constant $c > 0$. Define

$$P_1 = \frac{1}{K} + \varepsilon, \quad P_2 = \frac{1}{K} - \varepsilon, \quad P_k = \frac{1}{K} \quad (k \geq 3), \quad (20)$$

$$Q_1 = \frac{1}{K} - \varepsilon, \quad Q_2 = \frac{1}{K} + \varepsilon, \quad Q_k = \frac{1}{K} \quad (k \geq 3). \quad (21)$$

For large enough K , both distributions are valid and lie in an $O(1/K)$ neighborhood of u . Their entropy satisfies $H(P) = H(Q) = \log K - O(K\varepsilon^2) = \log K - O(1/K^2)$, so $\kappa = -\log K + O(1/K^2)$ and therefore $e^{-2\kappa} = \Theta(K^2)$.

Consider two point-mass models, θ_1 assigning all mass to action 1 and θ_2 assigning all mass to action 2. For any distribution R , the total variation error of the point mass at action j is

$$D_{\text{TV}}(\delta_j, R) = 1 - R_j. \quad (22)$$

Hence, under P , $D_{\text{TV}}(\delta_1, P) = 1 - P_1 < 1 - P_2 = D_{\text{TV}}(\delta_2, P)$, so θ_1 is better than θ_2 . Under Q the inequality is reversed. Any procedure that identifies the correct ranking in this local problem therefore distinguishes whether the samples came from P or Q .

It remains to bound the statistical distance between the two hypotheses. Let $a = 1/K$ and $t = \varepsilon/a = c/\sqrt{K}$. The per-sample KL divergence is

$$D_{\text{KL}}(P\|Q) = (a + \varepsilon) \log \frac{a + \varepsilon}{a - \varepsilon} + (a - \varepsilon) \log \frac{a - \varepsilon}{a + \varepsilon} \quad (23)$$

$$= 2\varepsilon \log \frac{1+t}{1-t}. \quad (24)$$

Since $t < 1/2$ for large enough K , $\log((1+t)/(1-t)) \leq Ct$ for a universal constant C , and thus

$$D_{\text{KL}}(P\|Q) \leq 2C \frac{\varepsilon^2}{a} = O(K\varepsilon^2) = O(1/K^2). \quad (25)$$

Let P^n and Q^n denote the n -sample product distributions. By tensorization and Pinsker's inequality,

$$D_{\text{TV}}(P^n, Q^n) \leq \sqrt{\frac{D_{\text{KL}}(P^n\|Q^n)}{2}} = \sqrt{\frac{n D_{\text{KL}}(P\|Q)}{2}} \leq C' \sqrt{\frac{n}{K^2}}. \quad (26)$$

Le Cam's lemma gives minimax error at least $\frac{1}{2}(1 - D_{\text{TV}}(P^n, Q^n))$. Therefore, if a ranking procedure achieves error probability at most $\frac{1}{2} - \eta$ for fixed $\eta > 0$, then $D_{\text{TV}}(P^n, Q^n) \geq 2\eta$, which requires $n \geq c_\eta K^2$. Since $e^{-2\kappa} = \Theta(K^2)$ in this construction, the required sample size is $\Omega(e^{-2\kappa})$. $\square$

C.4 SUPPORTING LEMMAS FOR CONTEXT AGGREGATION

We first record the properties of the effective action count that motivate Definition 2.

Remark 1 (Properties of the effective action count). *Definition 2 attaches the effective action count to a context through $\phi^*(\cdot | x)$. The statements below apply it to other distributions as well, so throughout the appendix we write*

$$K_{\text{eff}}(\phi) := \left(\sum_{k=1}^K \sqrt{\phi(k)} \right)^2, \quad \phi \in \Delta(\mathcal{A}), \quad (27)$$

for the same expression evaluated at an arbitrary distribution, so that $K_{\text{eff}}^*(x) = K_{\text{eff}}(\phi^*(\cdot | x))$ and the model-based estimate of Equation (18) is $\hat{K}_{\text{eff}}(x) = K_{\text{eff}}(\phi_\theta(\cdot | x))$.

The range $1 \leq K_{\text{eff}}(\phi) \leq K$, with the two endpoints attained at a point mass and at the uniform distribution, is verified inside the proof in Section C.5. Beyond the range, K_{eff} is exactly the Rényi perplexity of order $1/2$, that is $K_{\text{eff}}(\phi) = e^{H_{1/2}(\phi)}$ where $H_{1/2}$ is the Rényi entropy of that order. Since Rényi entropies are non-increasing in their order, $H_{1/2}(\phi) \geq H(\phi)$ for the Shannon entropy H , and therefore

$$K_{\text{eff}}(\phi^*(\cdot | x)) \geq e^{H(\phi^*(\cdot | x))} = K^{s(x)}, \quad (28)$$

with $s(x)$ the normalized subjectivity of Equation (9). The effective action count is thus lower-bounded by K raised to the normalized subjectivity, so the two grow together as behavior becomes more diffuse. This is why K_{eff} , rather than any other dispersion summary, is a natural effective-action measure for the statistical term of Lemma 2: the quantity that controls the statistical error is governed by the same notion of subjectivity that drives the rest of the analysis.

Finally, K_{eff} is continuous on the simplex: for any $\phi, \psi \in \Delta(\mathcal{A})$, using $|\sqrt{a} - \sqrt{b}| \leq \sqrt{|a - b|}$ termwise and then Cauchy-Schwarz,

$$\left| \sqrt{K_{\text{eff}}(\phi)} - \sqrt{K_{\text{eff}}(\psi)} \right| \leq \sum_{k=1}^K \sqrt{|\phi(k) - \psi(k)|} \leq \sqrt{K \sum_{k=1}^K |\phi(k) - \psi(k)|} = \sqrt{2K \cdot D_{\text{TV}}(\phi, \psi)}. \quad (29)$$

Two distributions that are close in total variation therefore have comparable effective action counts.

The analysis rests on the following smoothness assumption.

Assumption 1 (L -Lipschitz behavioral distribution). *There exists $L > 0$ such that for all $x, x' \in \mathcal{X}$: $D_{\text{TV}}(\phi^*(\cdot | x), \phi^*(\cdot | x')) \leq L \cdot d(x, x')$.*

This assumption should be read as behavioral smoothness in the chosen representation space. It requires that, within a fixed action-space group, nearby persona-context embeddings induce similar response distributions; the ablations over fixed neighborhoods, global frequency labels, and radius scale test whether this approximation is useful in SUBJSIM.

Lemma 1 (Three-term decomposition). *For any context x , let $\bar{\phi}(\cdot | x) := \frac{1}{|\mathcal{N}(x)|} \sum_{x' \in \mathcal{N}(x)} \phi^*(\cdot | x')$ denote the population mean of ϕ^* within the neighborhood of x . Then:*

$$\begin{aligned} & D_{\text{TV}}(\phi_\theta(\cdot | x), \phi^*(\cdot | x)) \\ & \leq \underbrace{D_{\text{TV}}(\phi_\theta(\cdot | x), \hat{\phi}(\cdot | x))}_{\varepsilon_{\text{opt}}} + \underbrace{D_{\text{TV}}(\hat{\phi}(\cdot | x), \bar{\phi}(\cdot | x))}_{\varepsilon_{\text{stat}}} + \underbrace{D_{\text{TV}}(\bar{\phi}(\cdot | x), \phi^*(\cdot | x))}_{\varepsilon_{\text{bias}}}. \end{aligned} \quad (30)$$

Lemma 2 (Statistical and bias bounds). *Under Assumption 1, with neighborhood radius r :*

$$\mathbb{E}[\varepsilon_{\text{stat}}] \leq \frac{1}{2} \sqrt{\frac{K_{\text{eff}}}{|\mathcal{N}(x)|}}, \quad \varepsilon_{\text{bias}} \leq L \cdot r, \quad (31)$$

where $K_{\text{eff}} = K_{\text{eff}}(\bar{\phi}(\cdot | x))$ is the effective action count of the neighborhood-averaged population distribution $\bar{\phi}$ of Lemma 1. Under Assumption 1, $\bar{\phi}(\cdot | x)$ lies within Lr in total variation of $\phi^*(\cdot | x)$, so by (29) it differs from $K_{\text{eff}}^*(x)$ of Definition 2 only through the same smoothness that already controls the bias term; the main statements are written with $K_{\text{eff}}^*(x)$ for readability. As a worst case one may substitute $K_{\text{eff}} = K$.

Corollary 1 (Aggregation eliminates the structural error of standard training). *In the single-observation regime, standard SFT fits one hard label per context and therefore incurs the point-mass error $D_{\text{TV}}(\phi_\theta^{\text{SFT}}, \phi^*) \geq 1 - p_{\text{max}}$ at that context. Context-aggregation achieves error $\varepsilon_{\text{opt}} + (L^{d_X} K_{\text{eff}}^*(x)/n_g)^{1/(d_X+2)}$ up to constants; when optimization error is negligible, this upper bound tends to zero as $n_g \rightarrow \infty$.*

Remark 2 (Estimating the effective action count from the model). *The theorem is stated for the oracle effective action count of the local population distribution. The practical rule in Equation (18) replaces it with $\hat{K}_{\text{eff}}(x)$ computed from the current model output. If $\hat{K}_{\text{eff}}(x) \in [K_{\text{eff}}^*(x)/c, c \cdot K_{\text{eff}}^*(x)]$ for some constant $c \geq 1$, then, since $r(x)$ scales as $K_{\text{eff}}^{1/(d_X+2)}$, the resulting radius is within a factor $c^{1/(d_X+2)}$ of the oracle radius, and the optimized upper bound changes only by constants. The exponent makes this tolerance generous in practice: with $d_X = 28$, even $c = 2$ gives $2^{1/30} \approx 1.02$. Without this approximation, the model-based rule should be read as an oracle-motivated heuristic rather than a direct consequence of Theorem 1.*

The approximation is what the training objective is designed to deliver. Applying the continuity bound (29) with $\phi = \phi_\theta(\cdot | x)$ and $\psi = \bar{\phi}(\cdot | x)$, the gap between the model-based count and the oracle count is controlled by exactly the quantity the aggregation loss in Equation (14) minimizes. In the ideal case where the optimization error ε_{opt} and the statistical error of the soft label both vanish, $\phi_\theta(\cdot | x) \rightarrow \bar{\phi}(\cdot | x)$ and hence $\hat{K}_{\text{eff}}(x) \rightarrow K_{\text{eff}}(\bar{\phi}(\cdot | x))$, so the practical radius returns to the oracle radius.

C.5 PROOF OF LEMMA 2 (STATISTICAL AND BIAS BOUNDS)

Proof. Statistical bound. Write $n_x = |\mathcal{N}(x)|$ and, for $x' \in \mathcal{N}(x)$, $p_{x',k} = \phi^*(k | x')$. Each coordinate $\hat{\phi}_k$ is an average of independent Bernoulli variables with possibly different means $p_{x',k}$. Let $\bar{\phi}_k = n_x^{-1} \sum_{x' \in \mathcal{N}(x)} p_{x',k}$. Then

$$\text{Var}(\hat{\phi}_k) = \frac{1}{n_x^2} \sum_{x' \in \mathcal{N}(x)} p_{x',k}(1 - p_{x',k}) \leq \frac{\bar{\phi}_k}{n_x}. \quad (32)$$

By linearity of expectation and Jensen's inequality ($\mathbb{E}[|X|] \leq \sqrt{\mathbb{E}[X^2]}$):

$$\mathbb{E}[\varepsilon_{\text{stat}}] = \frac{1}{2} \sum_{k=1}^K \mathbb{E}[|\hat{\phi}_k - \bar{\phi}_k|] \leq \frac{1}{2} \sum_{k=1}^K \sqrt{\text{Var}(\hat{\phi}_k)} \quad (33)$$

$$\leq \frac{1}{2} \sum_{k=1}^K \sqrt{\frac{\bar{\phi}_k}{n_x}} = \frac{1}{2\sqrt{n_x}} \sum_{k=1}^K \sqrt{\bar{\phi}_k}. \quad (34)$$

By (27), $\sum_k \sqrt{\bar{\phi}_k} = \sqrt{K_{\text{eff}}(\bar{\phi}(\cdot | x))}$, so $\mathbb{E}[\varepsilon_{\text{stat}}] \leq \frac{1}{2} \sqrt{K_{\text{eff}}/n_x}$. Note that by Cauchy–Schwarz, $\sum_k \sqrt{\bar{\phi}_k} \leq \sqrt{K \sum_k \bar{\phi}_k} = \sqrt{K}$ (since $\sum_k \bar{\phi}_k = 1$), so $K_{\text{eff}} \leq K$ always holds. When $\bar{\phi}(\cdot | x)$ is a point mass, $\sum_k \sqrt{\bar{\phi}_k} = 1$ so $K_{\text{eff}} = 1$; when $\bar{\phi}(\cdot | x)$ is uniform, $\sum_k \sqrt{1/K} = \sqrt{K}$ so $K_{\text{eff}} = K$.

Bias bound. For the overlapping neighborhood used by SALT, every $x' \in \mathcal{N}(x)$ satisfies $d(x', x) \leq r$. By convexity of D_{TV} and Assumption 1:

$$\varepsilon_{\text{bias}} = D_{\text{TV}}(\bar{\phi}(\cdot | x), \phi^*(\cdot | x)) \quad (35)$$

$$\leq \frac{1}{|\mathcal{N}(x)|} \sum_{x' \in \mathcal{N}(x)} D_{\text{TV}}(\phi^*(\cdot | x'), \phi^*(\cdot | x)) \quad (36)$$

$$\leq \frac{1}{|\mathcal{N}(x)|} \sum_{x' \in \mathcal{N}(x)} L \cdot d(x', x) \leq Lr. \quad (37)$$

□

C.6 PROOF OF THEOREM 1 (AGGREGATION–ESTIMATION TRADEOFF)

Proof. Substituting $|\mathcal{N}(x)| \asymp n_g r^{d_{\mathcal{X}}}$ into Lemma 2 and combining via Lemma 1 gives

$$\mathbb{E}[D_{\text{TV}}] \lesssim \varepsilon_{\text{opt}} + Lr + \sqrt{\frac{K_{\text{eff}}}{n_g r^{d_{\mathcal{X}}}}}. \quad (38)$$

The optimization term is independent of the neighborhood radius in this tradeoff. The bias term increases in r and the statistical term decreases in r . To find the optimal r , we differentiate with respect to r and set the result to zero:

$$L = \frac{d_{\mathcal{X}}}{2} \cdot \frac{1}{r} \cdot \sqrt{\frac{K_{\text{eff}}}{n_g r^{d_{\mathcal{X}}}}}, \quad (39)$$

which gives $L^2 r^{d_{\mathcal{X}}+2} \asymp K_{\text{eff}}/n_g$, yielding

$$r^* \asymp \left(\frac{K_{\text{eff}}}{n_g L^2} \right)^{1/(d_{\mathcal{X}}+2)}. \quad (40)$$

Substituting r^* back: the bias term is $Lr^* = L \cdot (K_{\text{eff}}/(n_g L^2))^{1/(d_{\mathcal{X}}+2)} = (L^{d_{\mathcal{X}}} K_{\text{eff}}/n_g)^{1/(d_{\mathcal{X}}+2)}$, and the statistical term is of the same order, giving the optimized upper bound in (17). □

D SALT IMPLEMENTATION DETAILS

On SUBSIM, SALT groups contexts by survey question, embeds each context with Qwen3-embedding-8b, retrieves adaptive-radius neighbors within the group, and trains on the resulting soft labels. The remainder of this section specifies each of these steps.

Context embeddings. We use Qwen3-embedding-8b (Zhang et al., 2025) as the context encoder, which produces 4096-dimensional vectors for Chinese text. Embeddings are precomputed once before training and cached on disk; they are not updated during training. All contexts within the same action-space group share the same situational description s . As a result, the embeddings primarily capture persona similarity within each group rather than situational variation.

Neighborhood Construction Pairwise distances between context embeddings are computed using the Euclidean (ℓ_2) distance. The full distance matrix is precomputed once at training startup using SciPy on CPU and cached for reuse. We estimate the intrinsic dimension $d_{\mathcal{X}}$ of the persona embedding space via PCA with a variance threshold of 0.90, yielding $d_{\mathcal{X}} = 28$. The adaptive radius $r(x)$ is then computed per context according to Equation 18, with scale factor $C = 0.2$.

Soft-label refresh schedule. Soft labels are refreshed every 30 training steps using the current model checkpoint. Per-epoch refresh adapts slowly in early training, whereas per-step refresh is computationally prohibitive and unstable. The 30-step interval balances label responsiveness with training efficiency.

Algorithm 1 Context-Aggregation Training with Adaptive Radius

Require: Dataset $\mathcal{D}$; pretrained context encoder; radius function ρ ; refresh interval T ; number of epochs E

Ensure: Trained model ϕ_θ

- 1: **Action-space partitioning:** group all contexts by action space: $\mathcal{G}_g = \{x : \mathcal{A}(x) = \mathcal{A}_g\}$
 - 2: Embed all contexts using the pretrained encoder
 - 3: Initialize ϕ_θ from a pretrained LLM
 - 4: **for** epoch = 1, ..., E **do**
 - 5: **for** each optimizer step **do**
 - 6: **if** the step index is a multiple of T **then**
 - 7: **for** each context x **do**
 - 8: Compute adaptive radius $r(x) \leftarrow \rho(\phi_\theta, x)$
 - 9: Retrieve neighborhood $\mathcal{N}(x) \leftarrow \{x' \in \mathcal{G}_g : d(x', x) \leq r(x)\}$
 - 10: Construct soft label $\hat{\phi}(k | x) \leftarrow |\mathcal{N}(x)|^{-1} \sum_{x' \in \mathcal{N}(x)} \mathbf{1}[a(x') = a^{(k)}]$
 - 11: **end for**
 - 12: **end if**
 - 13: Update θ on the current batch by minimizing $D_{\text{KL}}(\hat{\phi}(\cdot | x) \| \phi_\theta(\cdot | x))$
 - 14: **end for**
 - 15: **end for**
-

Table 2: Mapping between theoretical concepts, notation, and their SubjSim implementation.

Concept	Notation	SubjSim realization	Protocol role
Decision context	$x = (u, s)$	Annotator persona paired with a survey question; each persona has 30 demographic attributes.	Input to training and evaluation.
Action space	$\mathcal{A}(x)$	Candidate response options for the survey question.	Common support for distributional evaluation within each question.
Latent response-propensity target	$\phi^*(\cdot x)$	Probability-ball empirical distribution $\hat{\phi}_u^{\text{ball}}(\cdot x)$, used as an elicited distributional proxy rather than a repeated-choice frequency.	Hidden during training; used only as the evaluation target.
Hard observation	$a(x)$	Modal response under $\hat{\phi}_u^{\text{ball}}(\cdot x)$.	Hard-label proxy for SFT, DPO, PPO, and SALT targets.
Action-space group	$\mathcal{G}_g$	Persona-question contexts from the same survey question.	Restricts aggregation to comparable candidate options.
SALT neighborhood	$\mathcal{N}(x)$	Nearest contexts within the adaptive embedding radius, using Qwen3-embedding-8b embeddings.	Defines which hard observations SALT pools.
Effective action count (oracle)	$K_{\text{eff}}^*(x)$	Effective action count of the true response distribution at the context; never observed.	Appears in the bound of Theorem 1 and in the oracle radius.
Effective action count (model-based)	$\hat{K}_{\text{eff}}(x)$	The same functional applied to the model’s own distribution, $K_{\text{eff}}(\phi_\theta(\cdot x))$.	Sets the adaptive radius without using the hidden target.
Global baseline	Global-Freq	Question-level frequency of hard labels.	Ablation that removes persona conditioning.

Training hyperparameters. Table 3 reports the main training and method-specific hyperparameters for each method. All methods are trained on 8 GPUs; SALT uses per-device batch size 1 with 16 gradient-accumulation steps. Both the DPO and PPO policies are initialized from the SFT checkpoint. SALT uses a maximum sequence length of 1024 for evaluation (`ca_eval_max_length`).

Reward Model Construction (PPO). The reward model is initialized from the SFT checkpoint. For each context, the action with the highest probability in the annotator’s empirical distribution, i.e., the hard label defined in Section 5.1, is treated as the chosen response. Each remaining candidate is paired individually as a rejected response, yielding $K - 1$ preference pairs per context.

Table 3: Main training and method-specific hyperparameters for all methods.

Group	Parameter	Policy models					PPO aux.
		SALT	SFT	DPO	PPO	DSA	RM
General	Epochs	4	5	4	4	4	4
	Global batch size	128	128	96	128	128	128
	Learning rate	5e-6	5e-6	5e-7	1e-6	5e-6	1e-5
	LR schedule	Cosine	Cosine	Cosine	Cosine	Cosine	Cosine
	Warmup ratio	0.1	0.1	0.1	0.1	0.1	0.1
	Optimizer	AdamW	AdamW	AdamW	AdamW	AdamW	AdamW
	Max sequence length	1024	1024	1024	1024	1024	1024
DPO	β	—	—	0.07	—	—	—
	Loss type	—	—	Sigmoid	—	—	—
	Label smoothing	—	—	0.0	—	—	—
	FTX coefficient	—	—	0.0	—	—	—
PPO	ϵ_{clip}	—	—	—	0.2	—	—
	Target KL	—	—	—	6.0	—	—
	Initial KL coeff.	—	—	—	0.05	—	—
	Sampling temperature	—	—	—	0.7	—	—
	Top- p	—	—	—	0.9	—	—
SALT	Scale factor C	0.2	—	—	—	—	—
	PCA variance threshold	0.90	—	—	—	—	—
	Softmax temperature	1.0	—	—	—	—	—

Computational Cost. SALT and SFT are trained on eight NVIDIA H20 GPUs using DeepSpeed ZeRO-2. Context embeddings for all 16,500 training samples are precomputed in 40 seconds. Each training epoch takes approximately 35 minutes for SALT and 31 minutes for SFT.

E PER-DOMAIN RESULTS

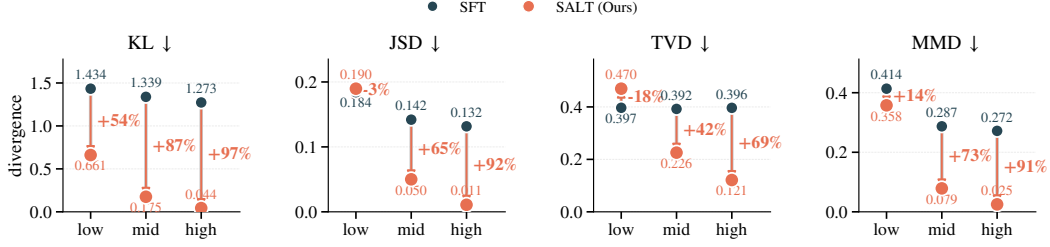
Figures 1 to 3 in the main text aggregate over the whole test set. This section repeats the same three analyses within each of the eight topic domains of SUBJ_{SIM}, so that the subjectivity trend, the neighbor-selection ablation, and the radius-scale sensitivity can each be checked domain by domain.

F STATISTICAL SIGNIFICANCE

Evaluation is deterministic (Section 5.2), so the only source of uncertainty in the reported metrics is the finiteness of the test set. We therefore quantify it with a paired bootstrap over test contexts: we resample the 2,800 test contexts with replacement 1,000 times, apply the same resampled index set to every method, and recompute each metric on every replicate. Table 4 reports the point estimates with 95% percentile intervals. The intervals of SALT and every baseline are disjoint on every metric; testing the paired differences directly, the 95% interval of SALT minus each comparator excludes zero on all four metrics, including the strongest ablation variant (fixed K_{eff} : KL difference -0.0273 , CI $[-0.0307, -0.0237]$). SALT’s improvements are therefore statistically significant rather than an artifact of the test split.

Table 4: Point estimates and 95% paired-bootstrap confidence intervals on the full test set (1,000 resamples over contexts). All metrics are divergences (lower is better).

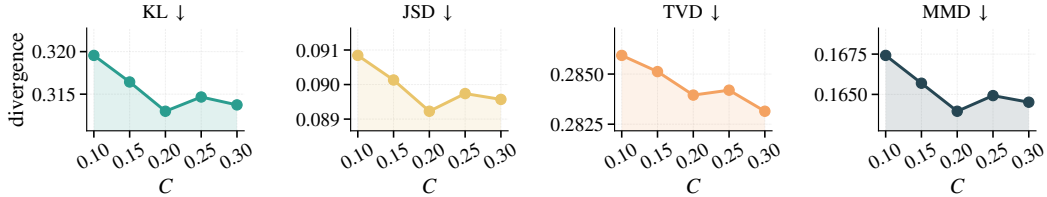
Method	KL		JSD		TVD		MMD	
Pretrained	4.2801	[4.1628, 4.3949]	0.2851	[0.2785, 0.2912]	0.5745	[0.5646, 0.5833]	0.6350	[0.6161, 0.6524]
SFT	1.2871	[1.2281, 1.3439]	0.1524	[0.1474, 0.1573]	0.3953	[0.3869, 0.4034]	0.3194	[0.3060, 0.3324]
DPO	6.8819	[6.7226, 7.0539]	0.2690	[0.2625, 0.2752]	0.5497	[0.5396, 0.5590]	0.5774	[0.5595, 0.5935]
PPO	4.1393	[4.0256, 4.2464]	0.2588	[0.2524, 0.2647]	0.5403	[0.5302, 0.5490]	0.5578	[0.5407, 0.5743]
DSA	3.3904	[3.2932, 3.4851]	0.2343	[0.2284, 0.2403]	0.4937	[0.4857, 0.5019]	0.4418	[0.4281, 0.4567]
SALT (Ours)	0.2880	[0.2749, 0.3009]	0.0825	[0.0788, 0.0863]	0.2720	[0.2653, 0.2787]	0.1510	[0.1440, 0.1582]
Top- N ($N=3$)	0.3653	[0.3497, 0.3809]	0.0925	[0.0886, 0.0964]	0.3078	[0.3011, 0.3144]	0.1921	[0.1833, 0.2014]
Fixed K_{eff} ($K=2$)	0.3152	[0.3017, 0.3290]	0.0900	[0.0861, 0.0939]	0.2839	[0.2769, 0.2909]	0.1656	[0.1586, 0.1734]
Global-Freq	0.3879	[0.3713, 0.4036]	0.0933	[0.0894, 0.0970]	0.3065	[0.2993, 0.3130]	0.1886	[0.1798, 0.1973]



(a) Subjectivity-stratified comparison between SFT and SALT

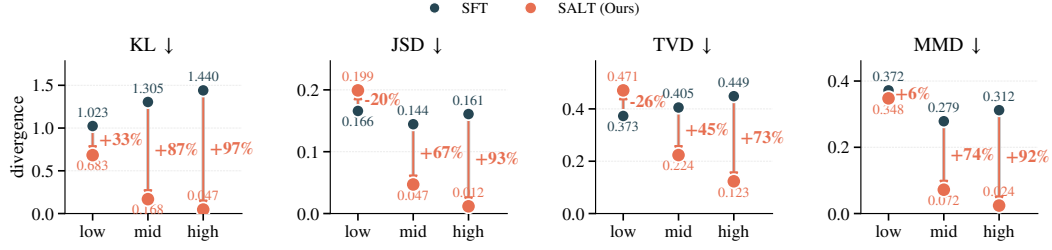


(b) Ablation over neighbor selection



(c) Sensitivity to the radius scale C

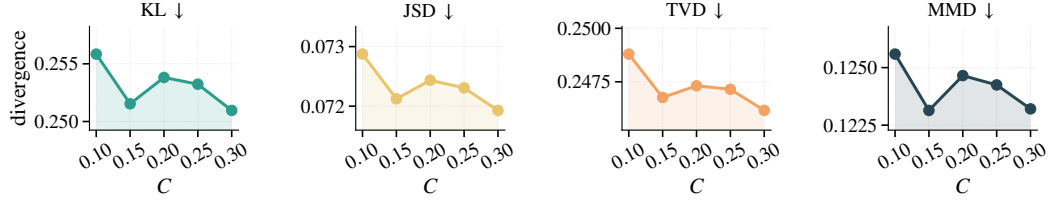
Figure 4: Per-domain results on the **Economy** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).



(a) Subjectivity-stratified comparison between SFT and SALT

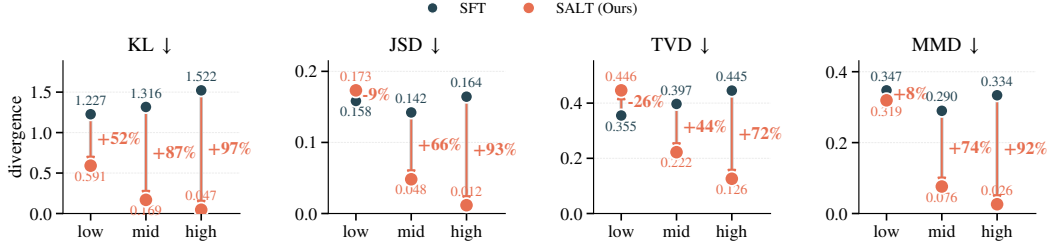


(b) Ablation over neighbor selection

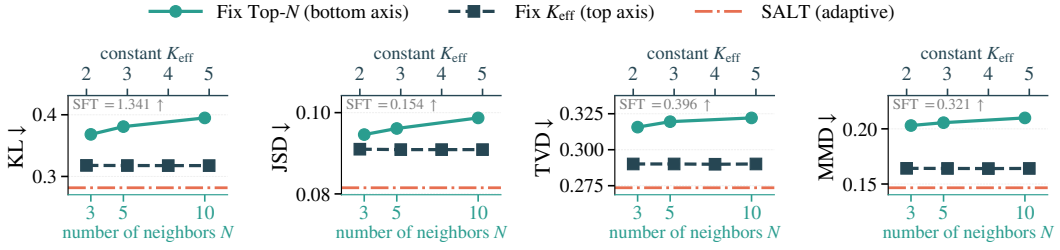


(c) Sensitivity to the radius scale C

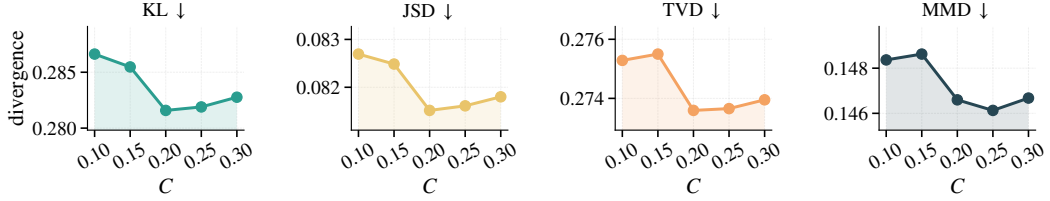
Figure 5: Per-domain results on the **Political** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).



(a) Subjectivity-stratified comparison between SFT and SALT

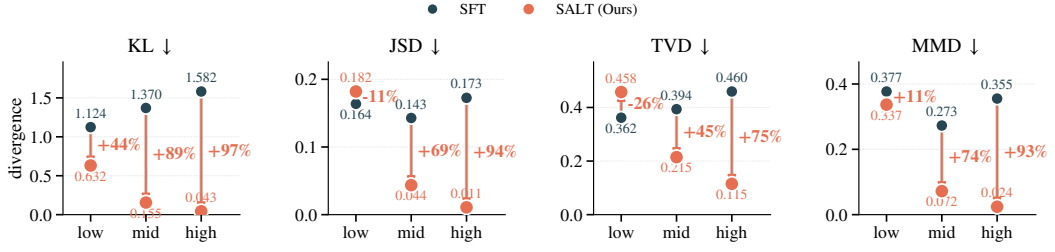


(b) Ablation over neighbor selection



(c) Sensitivity to the radius scale C

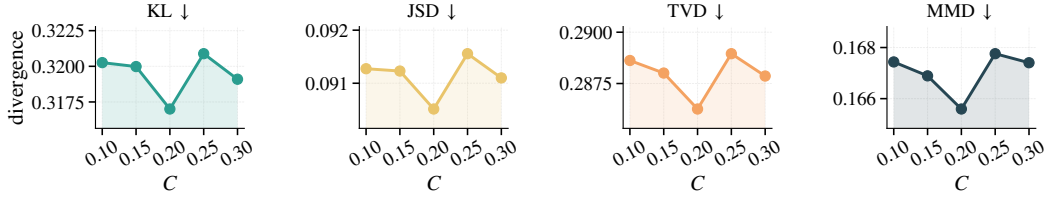
Figure 6: Per-domain results on the **Technology** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).



(a) Subjectivity-stratified comparison between SFT and SALT

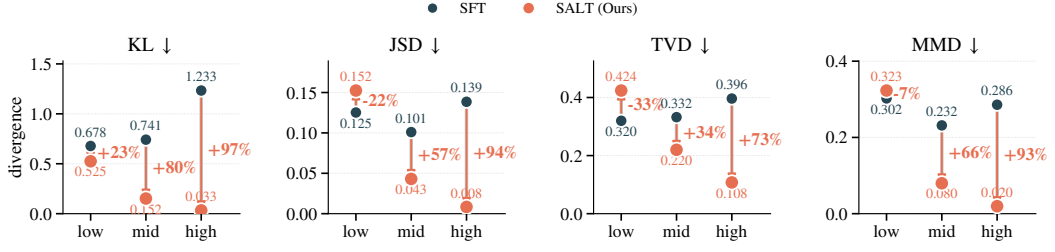


(b) Ablation over neighbor selection



(c) Sensitivity to the radius scale C

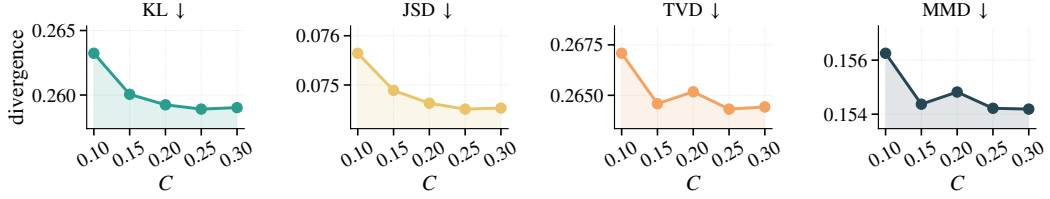
Figure 7: Per-domain results on the **Social** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).



(a) Subjectivity-stratified comparison between SFT and SALT

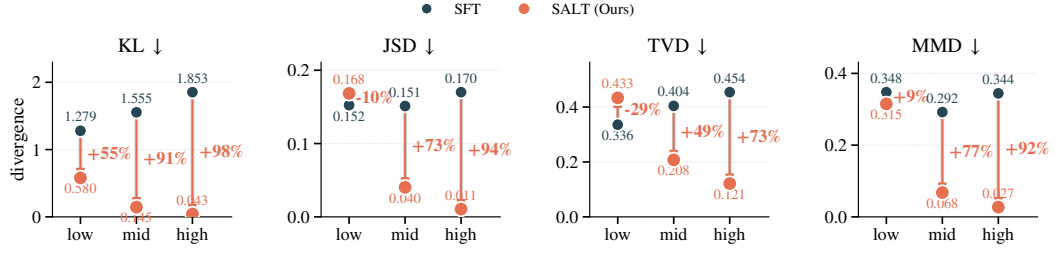


(b) Ablation over neighbor selection

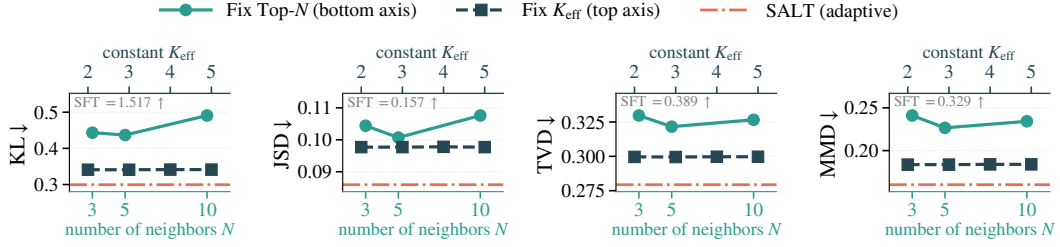


(c) Sensitivity to the radius scale C

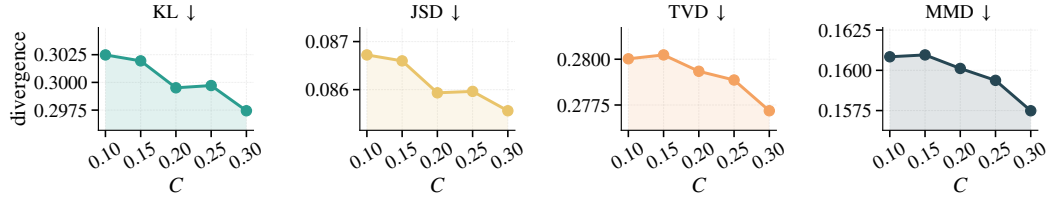
Figure 8: Per-domain results on the **Culture** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).



(a) Subjectivity-stratified comparison between SFT and SALT

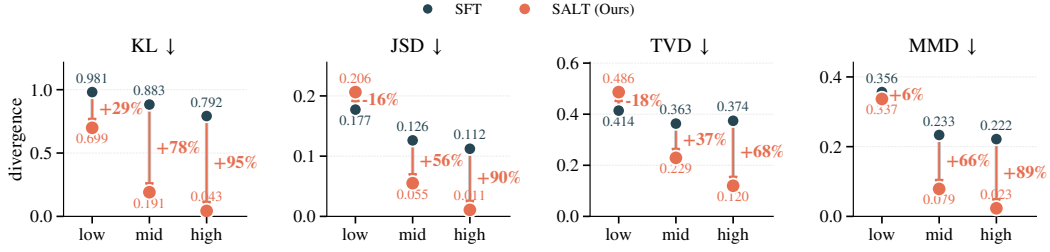


(b) Ablation over neighbor selection

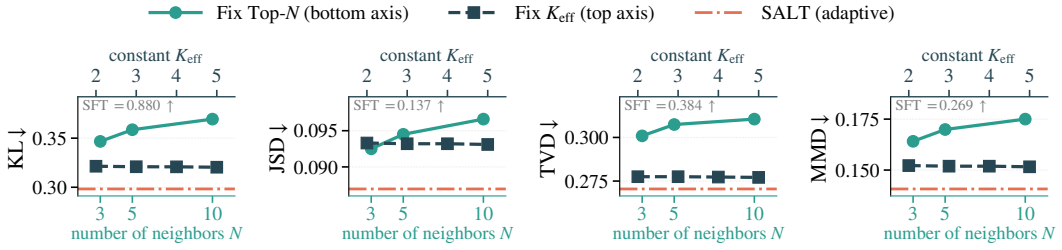


(c) Sensitivity to the radius scale C

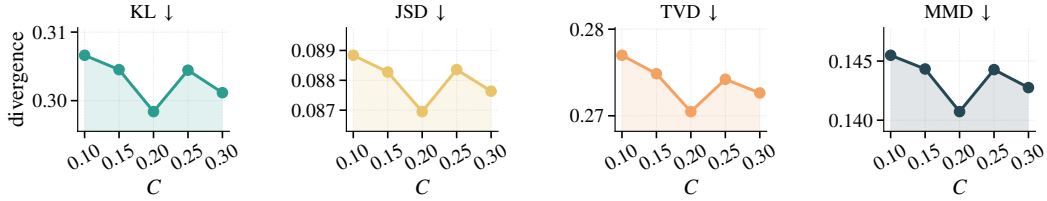
Figure 9: Per-domain results on the **Health** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).



(a) Subjectivity-stratified comparison between SFT and SALT



(b) Ablation over neighbor selection



(c) Sensitivity to the radius scale C

Figure 10: Per-domain results on the **Environment** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).

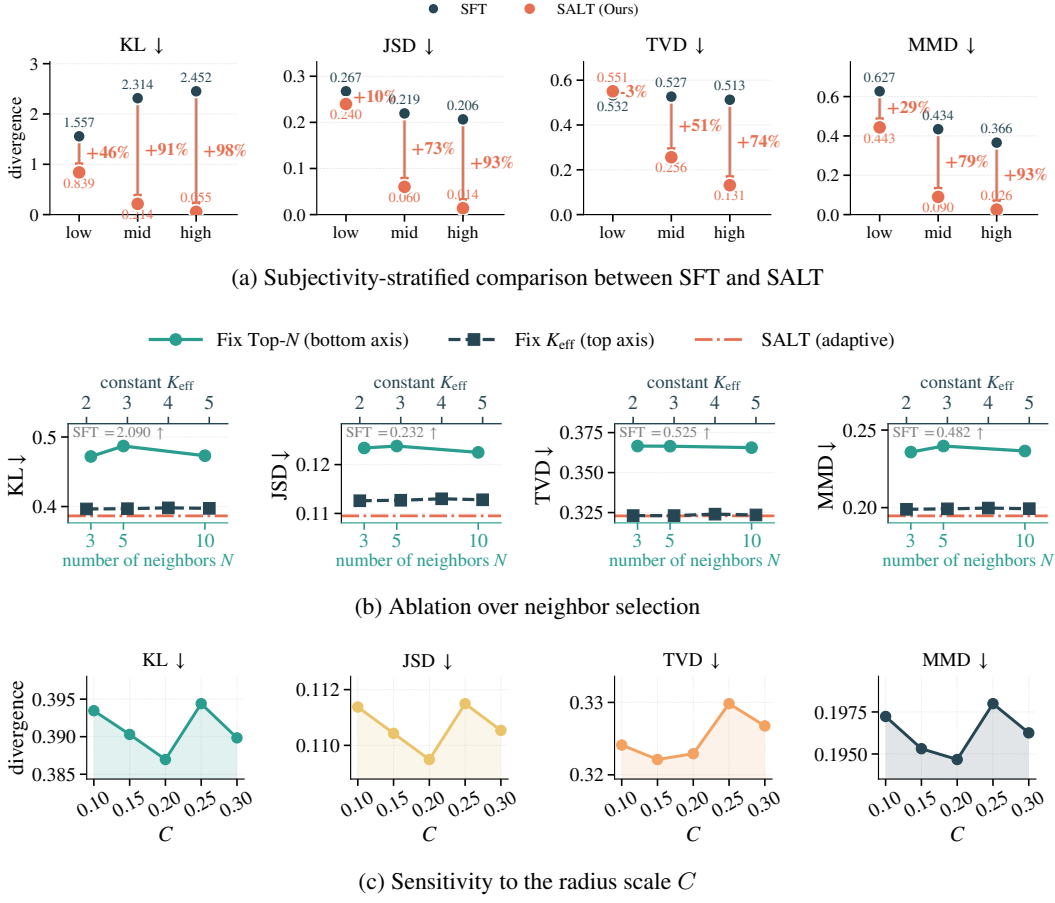


Figure 11: Per-domain results on the **Education** domain, mirroring Figure 1, Figure 2, and Figure 3 of the main text. All metrics are divergences (lower is better).

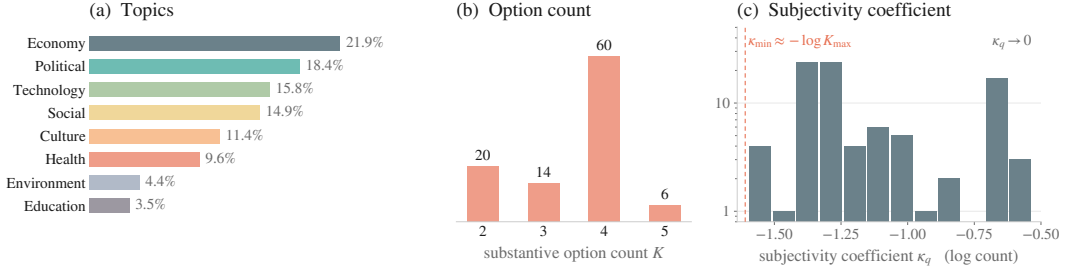


Figure 12: Overview of the SubjSim dataset. (a) Topics. (b) Option count K . (c) Subjectivity coefficient.

G SUBJSIM DATASET DETAILS

Why elicited propensities. Exact repeated measurement of the same person is not a viable route to the distributional ground truth, since repetition can change memory, reflection, fatigue, and demand effects. SUBJSIM therefore operationalizes the target via elicited subjective response propensities: respondents allocate plausibility across the same action space, so the hidden distributional target is available for evaluation while training methods receive only a single hard action.

G.1 QUESTION POOL AND FILTERING

Source Surveys. Questions are drawn from seven internationally standardized social survey programs: the American Trends Panel (ATP), General Social Survey (GSS), World Values Survey (WVS), American National Election Studies (ANES), Chinese General Social Survey (CGSS), European Social Survey (ESS), and International Social Survey Programme (ISSP). These programs were selected for their broad topical coverage, institutional authority, and diversity of question types. The initial candidate pool contains 5,964 items in total.

Translation. All non-Chinese items are translated into Chinese using DeepSeek-chat (Liu et al., 2024). CGSS items are retained in their original Chinese form.

Automated Screening. Each candidate item is evaluated by DeepSeek-chat (Liu et al., 2024) along three dimensions:

- **Cultural suitability:** whether the item is appropriate for Chinese respondents, considering cultural sensitivity, social norms, privacy boundaries, linguistic conventions, and regional variation.
- **Option ordinality:** whether the response options follow a logical order (ordinal) or are unordered categories (nominal). Only nominal items are retained, as ordinal scales introduce additional measurement assumptions that complicate distributional evaluation with metrics such as TVD and JSD.
- **Question objectivity:** whether the item has a factually verifiable answer (objective) or depends on personal attitudes, feelings, or preferences (subjective).

The system prompt used for automated screening is as follows:

Screening Prompt

You are a professional survey design expert. Please perform a three-dimensional annotation analysis on the given survey question.

Annotation Dimensions

- Cultural Suitability**
 - Determine whether the question is appropriate for Chinese respondents.
 - Consider: cultural sensitivity, social norms, privacy boundaries, linguistic conventions, and regional variation.

Table 5: Source distribution of the 100 subjective questions in SubjSim.

Source	Count	Proportion
ATP	35	35.0%
GSS	19	19.0%
WVS	14	14.0%
CGSS	9	9.0%
ANES	9	9.0%
ESS	9	9.0%
ISSP	5	5.0%
Total	100	100%

- Output: suitable / caution / unsuitable, with a brief explanation.

2. Option Ordinality

- Determine whether the response options have a logical order.
- Ordinal: options exhibit a clear gradient, ranking, or sequence (e.g., very dissatisfied → very satisfied; 18--25 → 26--35 → 36+).
- Nominal: options are parallel with no inherent order (e.g., red/blue/green; football/basketball/swimming).
- Output: ordinal / nominal, with justification.

3. Question Objectivity

- Determine whether the question is objective or subjective.
- Objective: the answer is factually verifiable and does not depend on personal feelings or opinions (e.g., age, household size, education level, occupation, home ownership, weekly exercise frequency).
- Subjective: the answer depends on personal attitudes, feelings, evaluations, or preferences (e.g., satisfaction, sense of identity, importance ratings, willingness, brand preference).
- Output: objective / subjective, with justification.

After automated screening, 109 items are labeled as suitable, nominal, and subjective. These items proceed to manual review. The remaining 1,012 suitable objective items serve as the pool for demographic attribute dimensions.

Manual Review and Localization. We manually review the 109 subjective items, remove duplicates, verify subjectivity, and adapt phrasing to the Chinese cultural context. This yields the final 100 questions.

Demographic Attribute Dimensions. From the 1,012 objective items, we manually select those most relevant to the 100 subjective questions, yielding 30 demographic attribute dimensions.

G.2 QUESTION AND DEMOGRAPHIC ATTRIBUTE EXAMPLES

Table 6: Examples of Demographic Attributes and Subjective Questions

Domain	Question	Response Options
Demographic Attributes		
Demographic	What is your current academic year?	1. Freshman/Sophomore; 2. Junior/Senior; 3. Master’s; 4. PhD; 5. Not a student
	Are you an only child?	1. Yes; 2. No
	What is your current employment status?	1. Full-time; 2. Part-time; 3. Self-employed; 4. Retired; 5. Homemaker; 6. Student; 7. Unemployed; 8. Other
Subjective Questions		
Political System	Over the next 30 years, which social trend concerns you the most?	1. AI replacing jobs; 2. Social stratification; 3. Misinformation; 4. Weakening family structures
	Among occupational groups, which do you trust the most?	1. Healthcare/education; 2. Law enforcement; 3. Business; 4. Non-profits
Education	Who should ensure young people acquire skills for good jobs?	1. Government; 2. Employers; 3. Education system; 4. Individuals
	Which quality is most important for children to learn?	1. Socially adept; 2. Obedient; 3. Hard work; 4. Helpful; 5. Independent thinking
Social Relations	What do you most often do in free time?	1. Social entertainment; 2. Leisure; 3. Self-improvement; 4. Exercise
	Which is more important: considerate or proper behavior?	1. Considerate; 2. Proper
Health Well-being	What is your view on vaccination?	1. Mandatory; 2. Personal choice; 3. Cautious; 4. Not mandatory
	What is the biggest problem with the healthcare system?	1. Over-prescription; 2. High cost; 3. Drug safety; 4. Uneven resources
Economy Labor	Do you prefer full-time employment?	1. Yes; 2. No
	What is the biggest challenge if changing jobs?	1. Salary/benefits; 2. Skill competitiveness; 3. Few opportunities; 4. Not difficult
Values Culture	Do you believe in life after death?	1. Yes; 2. No
	Which value should society prioritize?	1. Equal opportunity; 2. Individual freedom; 3. Social order; 4. Tradition
Technology & Society	What is your view on genetically modified foods?	1. Healthier; 2. More harmful; 3. No difference
	Who should protect personal information online?	1. Companies; 2. Individuals; 3. Public institutions
Environment & Energy	Which position do you lean toward on climate change?	1. Existential crisis; 2. Politicized; 3. Long-term issue; 4. Natural cycles
	Regarding energy, what are you most concerned about?	1. Prices; 2. Outages; 3. Fossil fuel reliance; 4. Natural disasters

G.3 ANNOTATOR DEMOGRAPHICS

We recruited 193 annotators via a university online forum, including both students and non-students. Figure 13 summarizes their distributions across age, gender, academic status, and field of study. The pool skews toward young, university-educated individuals, with a roughly balanced gender ratio and a mix of STEM and Humanities backgrounds. This limits claims about population-level

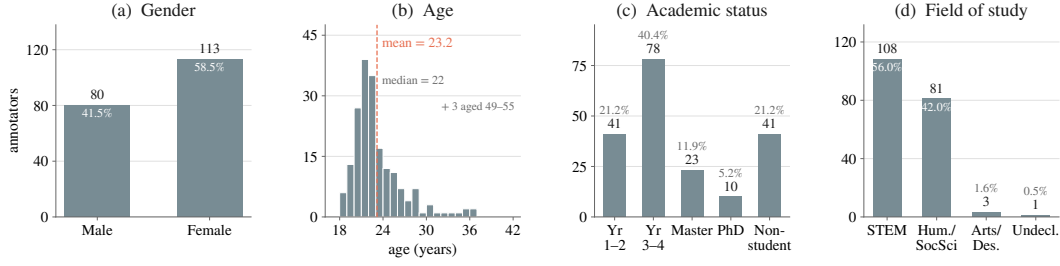


Figure 13: Demographic distributions of annotators across gender, age, academic status, and field of study.

representativeness, but it is less central to the benchmark’s main target: evaluating distributional alignment for individual behavioral tendencies.

G.4 PROBABILITY BALL ALLOCATION PROTOCOL

In the probability-ball protocol, annotators distribute a fixed number of balls across all available options and must allocate all balls before submission. The fraction assigned to each option represents the annotator’s subjective probability for that option. This avoids known issues with Likert-scale ratings such as scale bias and cross-item incomparability.

The number of balls scales with K to balance resolution and cognitive load: 10 balls for $K = 2$, 12 for $K = 3$ and $K = 4$, 15 for $K = 5$, and 20 otherwise.

We implemented a web-based annotation platform where annotators adjust ball counts per option via sliders or $+/-$ buttons. The remaining ball count is displayed in real time, and submission is blocked until all balls are allocated. Figure 14 shows a screenshot of the interface.

Question 28 / 60

Attitude DistributionATP

Allocation Rules

- Please allocate all 12 balls among the options.
- The more balls allocated, the closer the option is to your viewpoint.
- You can only submit and move to the next question when "Remaining Balls = 0".

Tip: Drag the slider for quick allocation, or click the '+' / '-' buttons for fine-tuning.

If gene editing technology is widely used to change the genetic characteristics of infants, what is your biggest concern?

Remaining Balls
12 / 12 Balls

1. Technology is misused for morally unacceptable purposes.

-0+

2. Only the wealthy can afford it, exacerbating social inequality.

-0+

3. The long-term effects on human health remain unclear.

-0+

4. This is the direction of scientific progress; I am not very concerned.

-0+

Next Question →

Figure 14: Screenshot of the annotation interface. The platform was deployed in Chinese for native Chinese-speaking annotators; the interface shown here is an English translation for presentation purposes.